



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2025-26 ISE2,
Class : TY Civil Engg.

Date : 15/10/2025
Time : 11.00 to 11.45

Subject: Hydropower Engineering
(Elective)
Code : PEC-CE305, 303 (Elective-1)

Max. Marks : 20
Duration: 45min

Assume suitable data if necessary.

Q1	A hydroelectric station is to be supplied with $10\text{m}^3/\text{s}$ of water through a penstock which has a friction factor of 0.016. The maximum normal head on the penstock is 50 kg/cm^2 and a water hammer overpressure of 20% is anticipated over normal pressure. The safe stress in the steel used is presumed to be 3000kg/cm^2 . The ready penstocks at the site are likely to cost Rs.65per kg including erection charges. The life of the project is 50 years and interest rate is 7%. It is proposed to sell the energy at the rate of Rs. 3.03 /kWh. What should be the optimum diameter of the penstock, given that the OMR cost are 5%. Assume the turbine efficiency to be 90% and annual load factor as 0.4.	10	CO3
Q.2	Derive an equation for efficiency of pure pumped-storage power plant and also state necessary assumptions made.	5	CO2
Q3.	Explain working of double basin tidal power station with neat sketches.	5	CO2



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Civil Engineering ISE-II Examination Academic Year: 2025-26 (Semester I)

Class: T.Y. B. Tech. (Civil Engg.)

Course: Engg. Hydrology

Course Code: PCC- CE303

Date: 16/10/2025

Time: 11:00 A.M.- 11:45 A.M

Max. Marks: 20

Note: i. Assume suitable data whenever necessary and state it clearly. ii. Mobile phones and programmable calculators are strictly prohibited. iii. Exchange/Sharing of stationery, calculator etc. not allowed. iv. Figures to right indicate full marks. v. The symbols carry usual meaning.

Q. No.	Question	Marks	CO	BT
Q.1	Explain in brief: (a) Flow mass curve (b) Flow duration curve	10	CO3	BT2
Q.2	What are the assumptions made in unit hydrograph theory? State and explain in brief the factors affecting runoff hydrograph.	10	CO4	BT3



S. G. G. S. Institute of Engineering and Technology

Vishnupuri, Nanded-431606, India

Department of Civil Engineering

IN-SEMESTER EXAMINATION - II (ISE-II) - 2025-26 (Semester-I)

Date: Friday, October 17, 2025

Course: Environmental Engineering

Time: 11:00 - 11:45

Class: T.Y. Civil Engineering

Course Code: PCC-CE302

Maximum Marks: 20

- Note
1. Attempt all questions. Read question statement carefully.
 2. Use of non-programmable calculator is permitted
 3. Assume appropriate data, if necessary.

Que. No.	Question statement	Mark	CO	BT
01	Design a sewer running 0.7 times full at maximum discharge for a town provided with the separate system, serving a population 80,000 persons. The water supplied from the water works to the town is at a rate of 190 LPCD. The manning's $n = 0.013$ for the pipe material and permissible slope is 1 in 600. Variation of n with depth may be neglected. (for $d/D = 0.7$, $q/Q = 0.838$, $v/V = 1.12$)	05	CO4	BT4
02	Enlist the sampling methods and explain anyone with example.	05	CO5	BT3
03	A sample was collected from the water tap for detection of the pollution by sewage. The dissolved oxygen measured immediate after collection was 7 mg/L. A sample was kept in the BOD incubator at 20°C for 5 days. After 5 days dissolved oxygen was observed 2.5 mg/L. Determine the 5-days biochemical oxygen demand (BOD) for this sample. What would be ultimate BOD of the sample, if $K_d = 0.1$ /day.	05	CO5	BT4
04	Differentiate between combined and separate sewerage system.	05	CO4	BT2

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1 Identify the source of water and water demand.	BT1 - Remember
CO2 Understand the water treatment concept and methods.	BT2 - Understand
CO3 Apply water distribution processes and operation and maintenance of water supply.	BT3 - Apply
CO4 Analyse the sewage characteristics and design various sewage treatment plants.	BT4 - Analyse
CO5 Evaluate basic process designs of water and wastewater treatment plants collect, reduce, analyse, and evaluate basic water quality data.	BT5 - Evaluate
CO6 Understand solid waste management	BT6 - Create

* * * All the Very Best * * *



S. G. G. S. Institute of Engineering and Technology
Vishnupuri, Nanded-431606, India
Department of Civil Engineering

IN-SEMESTER EXAMINATION - II (ISE-II) - AY 2025-26 (Semester-V)

Date: Friday, 17-October-2025

Course: Bridge Engineering

Time: 11:00 - 11:45

Class: Third Year (Common)

Course Code: OE-IS-CE301

Maximum Marks: 20

- Note
1. Attempt all questions. Read question statement carefully.
 2. Use of non-programmable calculator is permitted
 3. Assume appropriate data, if necessary. Draw necessary diagrams wherever required.

Que.	Question statement	Mark	CO	BT																		
01	Define Natural Waterway, and Restricted Waterway	04	CO2	BT1																		
02	Explain scour depth in bridges.	04	CO2	BT2																		
03	Suggest the span for a multi-span bridge based on following data obtained through the field conditions and subsequent analysis. Mention the reason behind your decision on provision. <table><tr><td>Span (m)</td><td>4</td><td>8</td><td>12</td><td>15</td><td>16</td></tr><tr><td>Super-structure cost (Rs.)</td><td>250000</td><td>270000</td><td>285000</td><td>300500</td><td>394000</td></tr><tr><td>Sub-structure cost (Rs.)</td><td>350150</td><td>360505</td><td>300000</td><td>286800</td><td>295466</td></tr></table>	Span (m)	4	8	12	15	16	Super-structure cost (Rs.)	250000	270000	285000	300500	394000	Sub-structure cost (Rs.)	350150	360505	300000	286800	295466	05	CO5	BT4
Span (m)	4	8	12	15	16																	
Super-structure cost (Rs.)	250000	270000	285000	300500	394000																	
Sub-structure cost (Rs.)	350150	360505	300000	286800	295466																	
04	Suggest a foundation depth based on hydraulic calculation for the river bridge. The flow of river is 150 m ³ /s. The waterway to be provided is 60 m. Assume Ksf as 1.37. Mention the reason behind your decision on provision of foundation depth.	05	CO2	BT4																		

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1. Know the bridges and their classifications	BT1 - Remember
CO2. Comprehend bridge hydraulic parameters	BT2 - Understand
CO3. Know the components of bridges	BT3 - Apply
CO4. Interpret the loadings on bridges	BT4 - Analyse
CO5. Explain the construction steps for bridges	BT5 - Evaluate
CO6. Articulate the testing and strengthening of bridges	BT6 - Create

*** All the Very Best ***



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
ISE-II (Semester-I) AY: 2025-26

Class: Third Year B. Tech.

Branch: Civil Engineering

Course: Building Materials and Concrete Technology

Course Code: PCC-CE301

Date: 14-Oct-2025

Time: 45 Minutes (11:00 to 11:45)

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	What are the various effects of impurities, present in water, on properties of concrete?	04	3	1
Q. 2)	Write down and explain the positive and negative effects of air entrainment on the properties of concrete.	04	3	2
Q. 3)	Explain the various factors affecting the workability.	04	4	2
Q. 4)	Explain the slump test, with neat sketch, for measurement of workability of the concrete.	08	4	2

(Page 1 of 2)

iv. The symbols carry usual meaning.

v. All questions are compulsory

Q.N.		Marks	CO	BT
Q.1	a	4	CO3	BT2
Q.2	a	4	CO2	BT5
Q.3	a	6	CO3	BT4
Q.4	a	6	CO3	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall stress and strain fundamentals, Hooke's law, principal stresses, strain energy, and column buckling.
CO 2	Understand how load intensity, stress, strain, bending moment, and shear force influence structural member behaviour under different loads.
CO 3	Apply pure bending and torsion principles to calculate bending and shear stresses in beams and shafts.
CO 4	Analyse stress distribution in beam cross-sections, principal stresses and determine critical loads for column buckling using Euler's and Rankine's formulae.
CO 5	Evaluate and interpret shear force and bending moment diagrams for complex beam configurations and evaluate the combined effect of bending and torsion.